

Lab 7: The Moon

Part I— Rules of Thumb

Calibrate your hand to measure angles

The apparent positions and separations of objects in the sky are not determined by the linear distances between two objects but by their angular separation. For example, the angular distance from the horizon to directly overhead, the zenith, is 90° . Not only can we measure the angular separation between two objects, but we can also measure the angular size of an object such as the Moon, the Sun, and galaxies. Sextants allow the precise measurement of angles, but not all of us carry one in our back pocket (plus they take some practice). Luckily, there is a *handy* way to measure angular size.

Hold your arm out straight in front of you. Now extend your thumb like you're hitchhiking. This will be one of your basic measuring devices. Now make a loose fist. There's your other device.

Record all your measurements and your calculations in your notebook

Measure the width of your thumb across the nail. Use the string to measure the distance from your eye to your thumb at arm's length. Get as accurate a measurement as possible. Calculate the angle subtended by your thumb in degrees.

Repeat this process for your fist (across the knuckles)

You now have two "devices" for estimating angles that are always with you. Use both of them to measure the angular size of a 2 things in the room. Pick 1 not-too-big object and measure its angular size with both your fist and your thumb. Then measure the distance from you to the object, and its actual size. Use these to calculate the angular size as well, and check how close your measurements are to reality. Why might your measurement be inaccurate?

Part II— The Moon

Draw a Portrait of the Moon

When we get up on the roof, Make two sketches of the Moon in Figure 1:

- One with a large field of view, so the whole disk is visible
- One with a small field of view, focusing on one section. Indicate on the larger drawing where you zoomed in for the close up drawing.

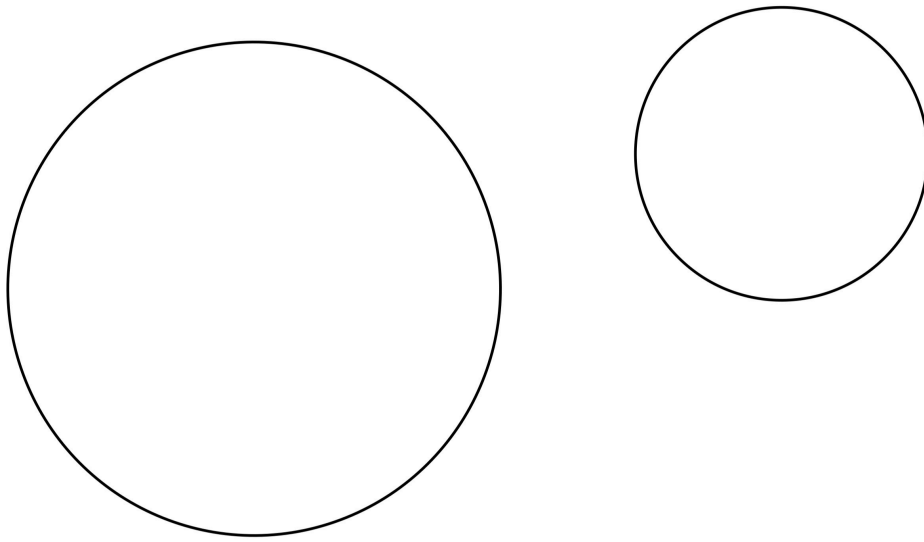


Figure 1: Your sketch of the Moon. The circle on the left is for the large field of view, and the circle on the right is the smaller field of view. Draw lines to indicate what region of the Moon you are zooming in on.

Also answer the following questions while you are on the roof or after:

1. Note the time. How many degrees above the horizon is the Moon? Use your fists to measure.
2. How many degrees across is the Moon? Use your thumb to measure.
3. What is the radius of the Moon if it is 384,400 km away?
4. What do you notice about the shadows? Can you find any craters with very large shadows compared to the size of the crater? What does this say about the height of the crater edge?
5. What kind of Moon is it? (i.e. Full, New, Waxing Gibbous? There is only one correct answer) Where is the sun? Draw a picture if necessary.

Your own picture of the Moon!

The telescope in the little dome is set up with a digital SLR camera to take pictures of the Moon. The telescope tracks with the sky so your pictures won't appear blurry. Select one region of the Moon (maybe the part which you focused on in your drawing, or at the terminator, or the edge are all interesting) and take a picture of it.

We will note down which picture is yours and e-mail you your own Moon picture later this week!

Part III — Phases of the Moon

For this activity, you will work in pairs and in the dark. You will try to model the Sun-Earth-Moon system. Every group will be assigned a Sun (flashlight) and a Moon (a ping-pong ball). One student is sitting on a chair corresponding to the observer on the Earth (student-Earth). The other student will use a flashlight to shed light on the Earth-Moon system (student-Sun). The student holding the Moon (student-Moon) will rotate around the student-Earth counterclockwise until they complete a full circle. The student-Earth must follow the rotation of the student-Moon by turning their chair around and observing what they see on the Moon. Which part of your Moon is bright and which is dark?

1. In Figure 2, sketch the shape of the Moon (as seen from the Earth). Make to clearly mark which part of the Moon is dark and which part of the Moon is bright.
2. Can you recognize the phases of the Moon? Which configuration gives a Full Moon? Which gives a New Moon (when you cannot see the Moon)? Indicate them in your diagram.
3. It takes the Moon 28 days to go around once. Let day 1 be the day of the New Moon. Note in the diagram which days correspond to each phase.
4. It takes about 28 days for the Moon to orbit the Earth, but closer to 30 days for a new Moon to occur. Why does this happen?
5. What is the phase of the Moon during a solar eclipse? During a lunar eclipse? Why don't they happen every month? Hint: You need perfect alignment.

Part IV — The Far Side of the Moon

The far side of the Moon, sometimes called the “dark side of the Moon”, is the lunar hemisphere that permanently faces away from the Earth. It is therefore not visible from the Earth's surface. The far hemisphere was first photographed by the Soviet Luna 3 probe in 1959, and was first directly observed by human eyes when the Apollo 8 mission orbited the Moon in 1968. Some conspiracy theorists have alleged that some Apollo astronauts had seen UFOs on the far side of the Moon but were told to keep quiet about them. Let's see why we can't see this mysterious side of the Moon. We always see the same side of the Moon. Why? Now repeat the previous experiment, but one student is the Moon, the other the Earth and the last one is a far observer standing outside the circle the Moon makes around the Earth. As the Moon rotates around the Earth, the student-Moon should always face the Earth.

1. Stand outside of the circle your classmate-Moon makes around Earth. Observe your classmate. Does the Moon rotate? Do you see the same side all the time? If not, what portion of the circle does she have to walk before you see the same side?
2. Why do we always see the same side of the Moon? Think about how long it takes for the Moon to complete one orbit around the Earth and how long it takes for the Moon to rotate once about its axis.
3. Why does the far side of the Moon have more craters?

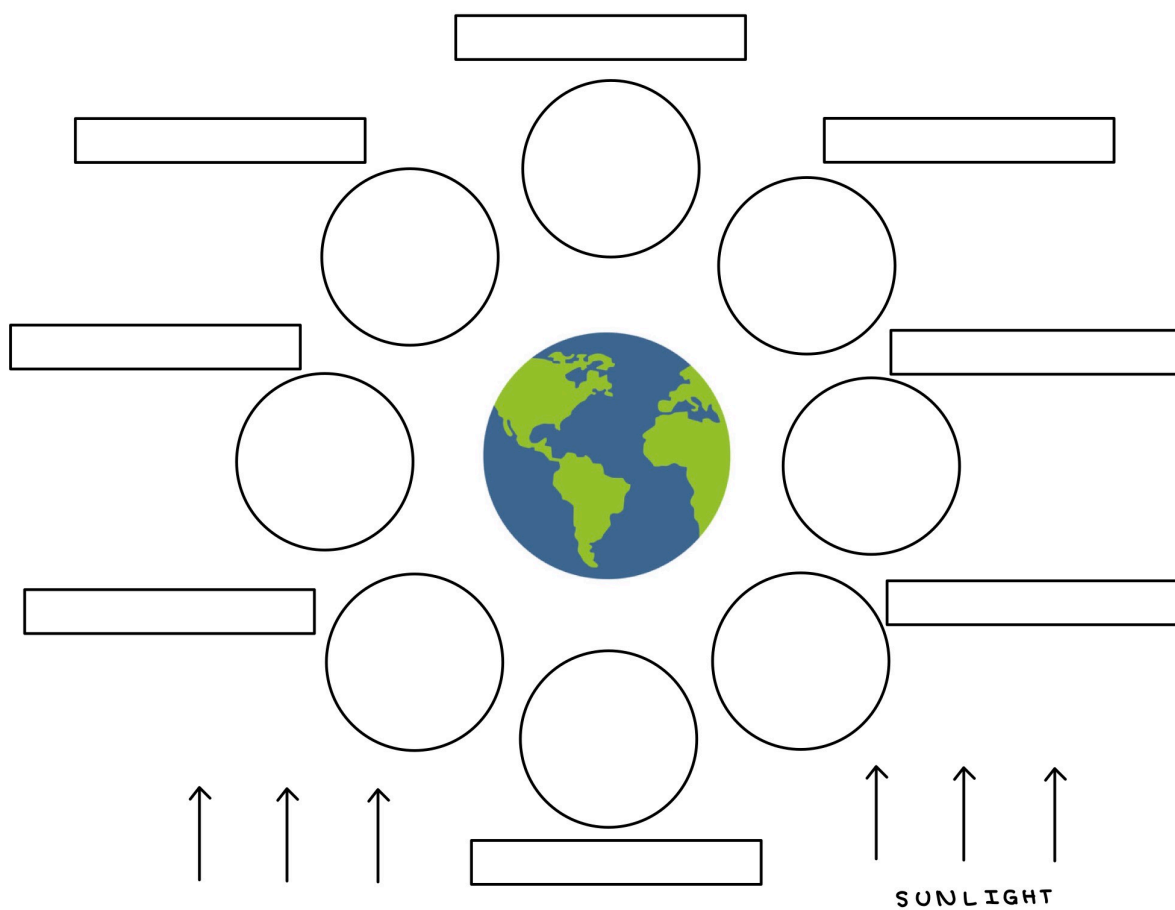


Figure 2: Phases of the Moon diagram. Sketch and label each phase of the Moon observed from Part III of the lab.

1 Part V — Tides

Both the Sun and the Moon exert gravitational force to the Earth. The Sun is more massive, but the Moon is much closer. Go to (Note: You have a 5 minute timer on the page; try using an incognito tab :D)

<https://gizmos.explorellearning.com/find-gizmos/lesson-info?resourceId=368>

Use the animation to understand how the gravitational pull of the Moon changes the level of the sea during the day. Explain which configuration of the Sun-Earth-Moon system gives the most extreme tides (highest high tide and lowest low tide). Sketch a diagram of the Sun-Earth-Moon system during the spring and neap tides.